

SNEHA MURALI

Sammamish, WA — U.S. Citizen

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🔗 github.com/snehamurali35

🌐 [Website Portfolio](#)

Education

University of Washington

Sep 2023 – June 2027

Bachelor of Science in Computer Science, Minor in Data Science

Seattle, WA

- **Relevant Coursework:** Data Structures, Software Design, Algorithms, Databases, Hardware/Software Interface, Mobile App Development, Computer Security, human computer interaction

Technical Skills

Languages: Java, Python, MySQL, PostgreSQL, R, JavaScript, Dart, C, C++

Developer Tools: React, Git, Firebase, Firestore, Firebase Authentication, HTML/CSS, Scikit-learn, Pandas, NumPy, Microsoft Excel, Figma, Tableau

Technologies/Frameworks: Flutter, Full-Stack Development, REST APIs, API Integration, Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Image Analysis Pipelines

Engineering Concepts : API Integration, Full-Stack Development, Role-Based Access Control (RBAC), Automation Workflows, Data Pipelines, Object-Oriented Programming, Database Migration, Backend Infrastructure

Experience

Comtech Telecommunications Corp. — Software Engineering Intern

June 2026 – Present

Seattle, WA

- Migrated the provisioning database for a Next Generation 911 (NG911) SMS provisioning system from MySQL to PostgreSQL using DBEaver, validating schema compatibility and supporting modernization of mission-critical emergency communications software.
- Upgraded Log4j dependencies and resolved application compatibility issues to improve the security, stability, and maintainability of the enterprise Java application.
- Developed full-stack enhancements to the NG911 SMS provisioning platform by extending SQL queries, backend services, and frontend UI components to display additional provisioning data like PSAP name, and ESINET name, to improve use.
- Extended an internal export tool by implementing backend logic to retrieve operational data from Splunk Cloud and load it into the WTP database, automating data integration for enterprise reporting workflows.

FridgeFinder

July 2025 – Present

Backend Developer

Remote

- Designed and implemented a DynamoDB-backed user rewards system in Python using Boto3, enabling atomic updates to user points and engagement metrics with concurrency-safe writes and strong consistency guarantees.
- Built and deployed scalable backend APIs using AWS Lambda and DynamoDB to aggregate, classify, and serve real-time community fridge status data for a production application (*fridgefinder.app*).
- Engineered robust error-handling and data validation logic to process incomplete, inconsistent, and unexpected real-world user reports, improving backend reliability and observability.
- Developed unit and integration tests using Moto to mock AWS services and validate DynamoDB operations, ensuring correctness and reliability of backend functionality.
- Collaborated in an Agile development environment using GitHub, contributing pull requests, code reviews, and iterative improvements to maintain production code quality.

UW Buckley Lab

March 2026 - Ongoing

Undergraduate Researcher

Seattle, WA

- Built Python pipelines to retrieve butterfly specimen images and associated metadata from museum and biodiversity APIs (iDigBio, GBIF, Smithsonian Open Access, Natural History Museum CKAN API), implementing API querying, nested JSON parsing, metadata standardization, and automated image downloading across large museum collections.
- Executed and validated Python-based image analysis pipelines to quantify butterfly coloration across large datasets.

- Implemented automated workflows for nested JSON parsing, metadata normalization, and large-batch image retrieval across heterogeneous museum data sources.
- Running, debugging, and monitoring terminal-based workflows; contributing to the development of automated data pipelines using Python and GitHub.

UW School of Medicine (SOM)

Feb 2026 – Ongoing

Web Developer Volunteer

Seattle, WA

- Built and maintained Firebase/Firestore-backed internal tools for volunteer event management, attendee tracking, and admin-facing workflows used by project coordinators.
- Developed role-based access systems using Firebase Auth and Firestore to securely manage participant information and permission-based admin views.
- Implemented backend data aggregation features to automate attendance tracking and improve visibility into volunteer engagement across events.
- Selected as Lead Developer for the 2026–2027 academic year to guide engineering priorities and mentor contributors.

Projects

Migraine Tracking App | *Flutter, APIs*

May – June 2025

- Developed a **cross-platform mobile app** using **Flutter** to track migraine symptoms and triggers.
- Integrated the **iOS Health API** and built **data visualizations** for symptom tracking.
- Built reusable UI components and implemented **state management**.

File System Search Engine | *C, C++, Data Structures, Systems Programming*

June - August 2025

- Built a **file system crawler, inverted index, and console-based search engine** in **C/C++** to recursively index and query text documents.
- Implemented core data structures including **hash tables, linked lists, iterators, and posting lists** to support efficient document indexing and ranked search.
- Engineered a **query processor** supporting multi-word search and relevance ranking based on document term frequency.
- Debugged memory and pointer issues using **Valgrind** and developed modular, testable systems code under strict performance and correctness requirements.

Traffic Congestion Optimization | *Python*

Dec 2023

- Developed a **neural network model** reducing congestion predictions by **15%**.
- Integrated external APIs and iterated under time constraints during a fast-paced hackathon environment.
- Awarded **Honorable Mention** at the **INRIX × Amazon Hackathon**.

Gestational Diabetes Prediction Model | *Python, Scikit-learn*

July – Aug 2024

- Developed a **logistic regression model** achieving **85% accuracy** for gestational diabetes prediction.
- Performed **feature engineering** and identified heredity as the strongest predictor.
- Evaluated model performance using **ROC curves, AUC**, and confusion matrices.

Music Match | *Python, AWS, PostgreSQL*

Jan – Mar 2024

- Built a **music recommendation system** using the **Spotify API, AWS EC2**, and **PostgreSQL**.
- Collaborated in **CSEED Cohort 0** to design and present a functional prototype.